

審查意見修改對照表

(2026-07-04；適用五份同步檔案：merged / merged_nowm / merged_3x / merged_ch / merged_ch_nowm)

□ 嚴重：Table 5.5 ↔ Table 5.17 矛盾 (FashionMNIST-source)

查證結果：逐格比對 Table 5.5 與其原始訓練 log (4090 / A6000 之 configC_cross 批次)：

- SVHN / STL10 / CINIC10 三個 source 的全部 12 列 (含 CINIC10 target 格) 與 log 逐格一致，無抄錄錯誤。
- FashionMNIST-source 的異常在來源批次本身：A6000 fashionmnist_configC_cross 的 SVHN M3 與 M4 兩個 log 之 Final Best Accuracy 為 45.68223724646589——到小數第 14 位完全相同。兩次獨立 200-epoch 訓練不可能如此，判定該批 FMNIST-source 存在 checkpoint / 評測重複問題，絕對值不可信。

處置(採審查建議 1：重跑)：已於 4090 (雙卡、4 路並行) 以完全相同之 configC 設定 (kernel 3 / ReLU / per-channel α / mid-basis out、200 epochs、seed 2024、RC) 重訓 FashionMNIST-source 全部 4 targets $\times$ M=1–4 (16 cells)。完成後將以新值整體置換 Table 5.5 之 FMNIST 四列 (含 Δ_{multi} 與粗體最大值)，屆時另出修訂版。本份對照表交付時重訓仍在進行中；除此四列外之所有修改已全部完成並重新編譯。

□ 中等

1. Backbone (Compact) 功耗 1.782 vs 1.747 不一致

Table 5.18 (tab:platform_baseline) compact 欄：

原文：Power (W) ...1.747；GOPs/W ...7.29；Efficiency ...61.3

改為：Power (W) ...1.782 (統一為 Table 5.13 之值)；GOPs/W ...7.14 (=12.73/1.782)；Efficiency ...60.1 (=107.1/1.782)

2. §5.8 限制條目 m2/m3 未跟上 §5.4.4

原文：m2/m3 hardware not yet built. The multi-adapter scaling curve ...is established in software. ...the projected accuracy gains have not been on-board-measured.

改為：m2/m3 not on-board verified. ...established in software and, at the hardware level, at the RTL-simulation and post-synthesis stages (Section 5.4.4): every multi-branch variant is bit-exact in per-MVAU RTL simulation and its synthesis-stage resource and power costs are measured. Place-and-route, however, is precluded by the XC7Z020's slice capacity (97.4% occupancy already at M=1), so on-board measurement of the multi-branch accuracy requires a larger FPGA (e.g., ZCU104, Alveo-class). (中文版同步改寫)

3. uniform PE = 1 與 Table 5.14 (FC3 = 5) 打架

三處修正 (§5.4.3、§5.6 開頭兩段；中文同步)：

原文：uses uniform PE = 1 across all MVAUs 改為：across all convolutional MVAUs (Conv1–Conv6)

原文：sets a uniform PE=1 in every MVAU—a deliberate choice **改為：**in every convolutional MVAU (Conv1–Conv6), the fully-connected layers retaining their original folding (FC3: PE = 5, Table 5.14)—a deliberate choice

原文：Reducing every MVAU to PE = 1 **改為：**Reducing every convolutional MVAU to PE = 1

4. 摘要（英+中）headline 更新為 16 對 mean/max

英文摘要：

原文：multi-adapter scaling raises 1-bit transfer accuracy by up to 6.1 percentage points over the single-branch baseline (most pronounced on SVHN and FashionMNIST)

改為：multi-adapter scaling raises 1-bit transfer accuracy over the single-branch baseline by an average of +3.5 percentage points across sixteen backbone–target pairs (up to +11.3 pp, most pronounced when transferring to SVHN)

中文摘要：**原文：**提升最高 6.1 個百分點（以 SVHN 與 FashionMNIST 最顯著）**改為：**於十六組 backbone–target 配對上平均提升 +3.5 個百分點（最高 +11.3，以遷移至 SVHN 最顯著）

§6 量化總結：於 CIFAR-10 之 +6.09/+3.12 後新增**改為：**「and across the full sixteen backbone–target sweep the mean multi-branch gain is +3.53 pp, up to +11.33 pp for FashionMNIST→SVHN (Section 5.4.4)」 （中文結論同步補句）。

5. 結論貢獻 #2 引用錯表

原文：(SVHN +6.09 pp, FashionMNIST +3.12 pp at M=4; Table 5.2)

改為：(...; Tables 5.11 and 5.17) (中文版「(表 5.2)」→「(表 5.11 與表 5.17)」)

6. Table 1.1 RQ3 證據欄

原文：Branch sweep (Section 5.3.3) **改為：**Branch sweep (Section 5.3.3; Table 5.17) (中文同步)

□ 敘述／文法／用詞

7. ”function multi-task switching” (11 處)

原文：function multi-task switching **改為：**runtime multi-task switching (全域替換 11 處，與 §5.6 標題 Runtime Multi-Task Dataset Switching 用語一致；中文版無此英文詞組、免改)

8. §2.3 LoRA 論證與文法

原文：per-task PEFT methods whose mechanism is to rewrite the backbone weights (e.g., the LoRA form $W+BA$) require regenerating the FINN memstreams ...; adapter-style PEFT instead inserts an additive side-stream ...

改為：per-task PEFT methods that, in their standard deployed (merged) form, fold the update into the backbone weights (the LoRA form $W+BA$) require regenerating the FINN memstreams ...; keeping the low-rank branch unmerged avoids the rewrite, but leaves a dense full-precision path with no mapping onto the XNOR–popcount MVAU primitive. Adapter-style PEFT instead inserts

an additive side-stream that leaves the backbone untouched and binarises onto the same streaming primitive (Chapter 3), integrating naturally via a side-FIFO + accumulator fusion (Section 4.3). (中文版同步改寫)

文法：**原文**：a property Transformer-style adapters or prompt-tuning do not— **改為**：...or prompt tuning do not preserve— (同時統一 prompt tuning 無連字號)

Conv-Adapter 數字：**原文**：approximately 3.5–5% ...across 23 cross-domain datasets **改為**：approximately 3.5% ...across 23 classification tasks (貼齊 [22] 原文；中文「約 3.5–5% ...23 個跨域資料集」→「約 3.5% ...23 個分類任務」)

9. §5.4.4 用詞

原文：every $M \geq 2$ configuration remains far beyond the device

改為：every $M \geq 2$ configuration remains beyond the device: still beyond on the compact build ($\approx 113\%$ LUT at $M=2$ after the same recovery) and far beyond on the throughput build

原文：the constraint anticipated in Section 5.4 **改為**：Section 5.4.1(新增 sec:resource_compare label 於「Cross-configuration resource comparison」小節，改指向該處)

10. 小點

- Figure 5.2(a) x 軸：查核確認 $M=1$ (90.1%) 之點在畫面內 (xlim 自 $\approx 78\%$ 起，僅刻度自 100 起跳)，不需改圖。
- Table 5.17 平均列： Δ_{multi} 欄 **原文**：+3.53 **改為**：— (避免與 $M=4$ 欄同值誤讀)。
- Jetson 能效：**原文**：143.16 **改為**：143.1 (表、內文、效率長條圖三處； $=2,752/19.23$ 之顯示捨入)。

狀態

除 Table 5.5 FMNIST 四列 (重訓中，另行交付) 外，上述所有修改已套用至五份檔案並重新編譯，undefined references = 0。另：CINIC10-backbone 之 deployed 幾何 $M=1-4$ 亦在 A6000 訓練中，完成後 Table 5.17 將由 16 對擴充為 20 對並同步更新圖 5.2/5.3 與摘要數字。